

Linear constraints

Elimination of variables

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Optimization: principles and algorithms



ÉCOLE POLYTECHNIQUE
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Elimination of constraints

We consider a polyhedron in standard form, with a matrix of full rank (no redundant constraint)

We use the linear equalities to eliminate variables

Example

$$\min x_1 + x_2 + x_3 + x_4$$

subject to

$$x_1 + x_2 + x_3 = 1$$

$$x_1 - x_2 + x_4 = 1$$

$$x_1, x_2, x_3, x_4 \geq 0.$$

$$x_3 = 1 - x_1 - x_2$$

$$x_4 = 1 - x_1 + x_2$$

$$\min x_1 + x_2 + 1 - x_1 - x_2 + 1 - x_1 + x_2 = -$$

Warning: $x_1 = 3, x_2 = 1, x_3 = -3,$
 $x_4 = -1$

Method

$$Ax = b, \quad A \in \mathbb{R}^{m \times n}, \quad b \in \mathbb{R}^m, \quad x \in \mathbb{R}^n, \quad \text{rank}(A) = m.$$

- ▶ Select m columns of A linearly independent to form $B \in \mathbb{R}^{m \times m}$:

$$AP = (B \ N) \text{ with } PP^T = I$$

- ▶ Rewrite the equality constraints:

$$Ax = (AP)(P^T x) = Bx_B + Nx_n = b$$

- ▶ Eliminate the basic variables:

$$x_B = B^{-1}(b - Nx_N).$$

Summary

Variables can be eliminated

Must choose m linearly independent columns of A

There are many combinations

Constraints on x_B are not eliminated